Worksheet 1B: Fractions in the Wild

A Note Before You Begin

This worksheet contains formulas from physics, machine learning, probability, and information theory. Some of them look intimidating. They're not.

You don't need to understand what these formulas *do* in their original context. You just need to look at them as fractions and ask: "What happens when this variable gets bigger? Smaller? Approaches some limit?"

This is about building confidence. If you can analyze the behavior of Einstein's equations, you can handle anything.

Keep calm. These are just fractions.

Part A: Physics — Special Relativity

In Einstein's special relativity, several important quantities involve the same fraction pattern.

The Lorentz Factor

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Here: - v = speed of an object - c = speed of light (the maximum possible speed) - The object cannot exceed c, so $v < c$ always

1. - (a) When $v = 0$ (object at rest), what is $\frac{v^2}{c^2}$? What is γ? - (b) When $v = 0.5c$ (half light speed), what is $\frac{v^2}{c^2}$? What is $1 - \frac{v^2}{c^2}$? - (c) As v approaches c, what does $\frac{v^2}{c^2}$ approach? - (d) As v approaches c, what does $1 - \frac{v^2}{c^2}$ approach? - (e) As v approaches c, what happens to γ?

Time Dilation

A moving clock runs slower. The relationship is:

$$\text{time_moving} = \text{time_stationary} \times \sqrt{1 - \frac{v^2}{c^2}}$$

2. - (a) When $v = 0$, how does time_moving compare to time_stationary? - (b) As $v \to c$, what happens to time_moving? - (c) In words: as you move faster, does your clock run faster or slower compared to a stationary observer?

Relativistic Energy

$$\text{energy} = \frac{\text{rest_mass} \times c^2}{\sqrt{1 - \frac{v^2}{c^2}}}$$

3. - (a) When $v = 0$, what is the energy? (This is the famous $E = mc^2$) - (b) As $v \to c$, what happens to the energy? - (c) Why does this suggest that nothing with mass can actually reach light speed?

Relativistic Momentum

$$\text{momentum} = \frac{\text{mass} \times v}{\sqrt{1 - \frac{v^2}{c^2}}}$$

4. - (a) When v is very small compared to c, the denominator is approximately 1. What does momentum approximately equal? - (b) As $v \to c$, what happens to momentum?

Part B: Probability — Bayes' Theorem

Bayes' theorem tells us how to update our beliefs when we get new evidence:

$$P(\text{hypothesis}|\text{evidence}) = \frac{P(\text{evidence}|\text{hypothesis}) \times P(\text{hypothesis})}{P(\text{evidence})}$$

This reads as: "The probability of our hypothesis *given* the evidence equals…"

Let's use a concrete example: **Medical Testing**

- `P(disease)` = probability someone has a disease (let's say it's rare: 1%)
- `P(positive | disease)` = probability of testing positive if you have the disease (sensitivity: 99%)
- `P(positive | no_disease)` = probability of testing positive if healthy (false positive rate: 5%)

5. The formula becomes:

$$P(\text{disease}|\text{positive}) = \frac{P(\text{positive}|\text{disease}) \times P(\text{disease})}{P(\text{positive})}$$

- (a) If the disease is very rare (small `P(disease)`), what happens to the numerator?
- (b) If the test has many false positives (high `P(positive | no_disease)`), what happens to `P(positive)` in the denominator?

- (c) Combining (a) and (b): even with a positive test, why might the probability of actually having a rare disease still be low?

6. Now imagine a very common condition where `P(condition) = 0.5` (50% of people have it). - (a) How does this change the numerator compared to a rare disease? - (b) Is a positive test more or less meaningful for common vs rare conditions?

7. What happens to `P(hypothesis | evidence)` if: - (a) `P(evidence)` is very small (the evidence is rare)? - (b) `P(evidence)` is close to 1 (the evidence is extremely common)?

Part C: Information Theory — Entropy and Compression

Shannon Entropy

The "surprise" or information content of an event is:

$$\text{surprise}(\text{event}) = \log_2\left(\frac{1}{P(\text{event})}\right)$$

8. - (a) If an event is very likely (`P(event)` close to 1), what is $\frac{1}{P(\text{event})}$ close to? - (b) So what happens to the surprise value? - (c) If an event is very unlikely (`P(event)` close to 0), what happens to $\frac{1}{P(\text{event})}$? - (d) So what happens to the surprise value? - (e) In plain English: are rare events more or less surprising?

9. Average entropy (average surprise) across all possible events:

$$H = \sum_{\text{each event } i} P(\text{event}_i) \times \log_2\left(\frac{1}{P(\text{event}_i)}\right)$$

Consider a coin flip: - Fair coin: `P(heads) = 0.5`, `P(tails) = 0.5` - Biased coin: `P(heads) = 0.99`, `P(tails) = 0.01`

- (a) Which coin is more predictable?
- (b) Which coin has lower entropy (less average surprise)?
- (c) Which coin's outcomes could be compressed more efficiently? Why?

Compression Ratio

$$\text{compression_ratio} = \frac{\text{compressed_size}}{\text{original_size}}$$

10. - (a) If compressed size is much smaller than original, is the ratio close to 0 or close to 1? - (b) What compression ratio means "no compression at all"? - (c) Can the compression ratio ever be greater than 1? What would that mean?

Part D: Machine Learning Concepts

The Sigmoid Function

The sigmoid "squashes" any input to a value between 0 and 1:

$$\text{sigmoid}(x) = \frac{1}{1 + e^{-x}}$$

Remember: $e \approx 2.718$, and $e^{-x} = \frac{1}{e^x}$

11. - (a) When $x = 0$, what is e^{-x}? What is sigmoid(0)? - (b) When x is very large and positive, what happens to e^{-x}? What does sigmoid approach? - (c) When x is very large and negative, what happens to e^{-x}? What does sigmoid approach? - (d) Can sigmoid ever actually equal 0 or 1 exactly?

12. The sigmoid is used to convert "confidence scores" to probabilities. Why is it useful that: - (a) The output is always between 0 and 1? - (b) An input of 0 gives output 0.5 (maximum uncertainty)?

Softmax Function

Softmax converts a list of numbers into probabilities that sum to 1:

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_{\text{all } j} e^{z_j}}$$

For example, with three scores $[z_1, z_2, z_3] = [2, 1, 0]$:

$$\text{softmax}(z_1) = \frac{e^2}{e^2 + e^1 + e^0}$$

13. - (a) What is the denominator ensuring about all the softmax outputs together? - (b) If one score z_i is much larger than the others, what happens to its softmax value? - (c) If all scores are equal, what are all the softmax values?

Learning Rate Decay

During training, we often shrink the learning rate over time:

$$\text{learning_rate}(t) = \frac{\text{initial_rate}}{1 + \text{decay} \times t}$$

Where t = time step (0, 1, 2, 3, …)

14. - (a) At $t = 0$ (start of training), what is the learning rate? - (b) As t gets very large, what does the learning rate approach? - (c) Why might we want the learning rate to start high and decrease over time?

Gradient Descent Update

When training a model, we update weights using:

$$\text{new_weight} = \text{old_weight} - \text{learning_rate} \times \frac{\text{change_in_loss}}{\text{change_in_weight}}$$

15. The fraction $\frac{\text{change_in_loss}}{\text{change_in_weight}}$ is called the gradient. - (a) If the gradient is positive (increasing the weight increases the loss), should we increase or decrease the weight? - (b) If the gradient is very large, should we take a big step or small step? - (c) Why do we multiply by learning rate — what does a small learning rate do to our steps?

Batch Normalization (Simplified)

$$\text{normalized} = \frac{\text{value} - \text{mean}}{\text{standard_deviation}}$$

16. - (a) If a value equals the mean, what is the normalized value? - (b) If a value is one standard deviation above the mean, what is normalized? - (c) What happens if standard deviation is very small (all values nearly identical)?

Part E: Exponential Growth and Decay

Compound Interest

$$\text{final_amount} = \text{principal} \times \left(1 + \frac{\text{rate}}{n}\right)^{n \times \text{years}}$$

Where n = number of times compounded per year

17. With rate = 100% (doubling) and 1 year: - (a) If $n = 1$ (yearly): $\left(1 + \frac{1}{1}\right)^{1} = ?$ - (b) If $n = 2$ (semi-annually): $\left(1 + \frac{1}{2}\right)^{2} = ?$ - (c) If $n = 12$ (monthly): $\left(1 + \frac{1}{12}\right)^{12} = ?$ (use calculator) - (d) If $n = 365$ (daily): $\left(1 + \frac{1}{365}\right)^{365} = ?$ (use calculator) - (e) As $n \to \infty$, this approaches $e \approx 2.718$. What does "continuous compounding" mean?

Radioactive Decay / Half-Life

$$\text{remaining} = \text{initial} \times \left(\frac{1}{2}\right)^{\frac{\text{time}}{\text{half_life}}}$$

18. - (a) After exactly one half-life, what fraction remains? - (b) After two half-lives? - (c) After ten half-lives? - (d) Does the amount ever reach exactly zero?

Model Accuracy Over Training

Imagine accuracy improves but with diminishing returns:

$$\text{accuracy}(t) = 1 - \frac{1}{1+t}$$

Where t = training epochs

19. - (a) At $t = 0$, what is accuracy? - (b) At $t = 1$? - (c) At $t = 9$? - (d) At $t = 99$? - (e) What value does accuracy approach but never reach?

Part F: Classification Metrics

Precision

$$\text{precision} = \frac{\text{true_positives}}{\text{true_positives} + \text{false_positives}}$$

"Of everything we predicted positive, how many were actually positive?"

20. - (a) If we have 90 true positives and 10 false positives, what is precision? - (b) If false positives increase while true positives stay the same, does precision go up or down? - (c) What precision value means "every positive prediction was correct"?

Recall

$$\text{recall} = \frac{\text{true_positives}}{\text{true_positives} + \text{false_negatives}}$$

"Of everything that was actually positive, how many did we catch?"

21. - (a) If we have 90 true positives and 10 false negatives, what is recall? - (b) If we predict *everything* as positive (to miss nothing), what happens to false negatives? What is recall? - (c) But what happens to precision if we predict everything as positive?

F1 Score

The F1 score balances precision and recall:

$$F1 = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

22. - (a) If precision = 1 and recall = 1, what is F1? - (b) If precision = 1 and recall = 0, what is F1? - (c) If precision = 0.6 and recall = 0.6, what is F1? - (d) This is called the "harmonic mean." Why might we prefer this over a simple average $\frac{\text{precision}+\text{recall}}{2}$?

Accuracy vs Class Imbalance

$$\text{accuracy} = \frac{\text{correct_predictions}}{\text{total_predictions}}$$

23. Suppose 99% of emails are not spam, and only 1% are spam. - (a) If a model predicts "not spam" for everything, what is its accuracy? - (b) How many spam emails does it catch? - (c) Why is accuracy misleading for imbalanced datasets?

Part G: Geometric Series and Convergence

24. The infinite geometric series formula:

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r} \quad (\text{when } |r| < 1)$$

- (a) Verify: $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + ... = \frac{1}{1-0.5} = ?$
- (b) What is $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + ...$?
- (c) What is $1 + \frac{2}{3} + \frac{4}{9} + \frac{8}{27} + ...$? (Here $r = \frac{2}{3}$)
- (d) Why doesn't this formula work when $r \geq 1$?

25. Expected value in probability uses similar patterns. If you have a $\frac{1}{2}$ chance of winning on each try:

$$\text{expected_tries} = \frac{1}{P(\text{success})} = \frac{1}{1/2} = 2$$

- (a) If success probability is $\frac{1}{6}$ (like rolling a specific number on a die), expected tries?
- (b) If success probability is $\frac{1}{100}$, expected tries?
- (c) As success probability approaches 0, expected tries approaches…?

Part H: Challenge Problems

26. The logistic growth model (populations, adoption curves):

$$P(t) = \frac{K}{1 + \left(\frac{K - P_0}{P_0}\right) e^{-rt}}$$

Where K = carrying capacity, P_0 = initial population, r = growth rate, t = time

Don't panic! Just analyze the fraction: - (a) As $t \to \infty$, what happens to e^{-rt}? - (b) As $t \to \infty$, what does the denominator approach? - (c) As $t \to \infty$, what does $P(t)$ approach? - (d) At $t = 0$, verify that $P(0) = P_0$.

27. The normal distribution's probability density:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \times e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Focus just on the exponent: $-\frac{(x-\mu)^2}{2\sigma^2}$

- (a) When $x = \mu$ (at the mean), what is the exponent? What is e^0?
- (b) As x moves far from μ, does $(x-\mu)^2$ get larger or smaller?
- (c) So does the exponent become more negative or less negative?
- (d) What happens to the overall probability density far from the mean?

28. In neural networks, the "attention" mechanism uses:

$$\text{attention_weight}_i = \frac{e^{\text{score}_i}}{\sum_j e^{\text{score}_j}}$$

This is just softmax! If one item has score 10 and two others have score 0: - (a) What is the attention weight for the high-score item (approximately)? - (b) What are the attention weights for the low-score items? - (c) What does "attention" mean in plain English here?
